The Box Principle — problem 4.78

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Let $a_1, \ldots, a_{100}$ and $b_1, \ldots, b_{100}$ be two permutations of the integers from 1 to 100. Prove that, among the products $a_1b_1, \ldots, a_{100}b_{100}$, there are two with the same remainder upon division by 100.